

Regime-Switching Asset Allocation via a POMDP

Type: New Application · 12 baselines, 5 diagnostic experiments

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The problem: regimes break static allocation

In 2022 a 60/40 portfolio lost $\sim 17\%$.

Static allocation assumes stable joint return moments. Markets violate this in two ways:

- **Conditional heteroskedasticity:** long quiet periods broken by short crises (2008,

Question. Can a decision-theoretic regime overlay—a POMDP solved end-to-end on macroeconomic observables—beat a static benchmark after realistic frictions?

Why POMDP, not a tree, ID, or fully-observable MDP

Naive choice	Why it fails here
Decision tree	Tree explodes combinatorially with horizon; no shared state.
Influence diagram	Same blow-up; not natural for repeated monthly rebalance.
Fully-observable MDP	Requires knowing the regime — but it is <i>latent</i> .
POMDP	Hidden state, Bayesian belief, QMDP for tractable control.

Powell (2019) taxonomy: QMDP is a **value function approximation** (linear lower envelope over the belief simplex) combined with a **one-step direct lookahead**. The right *minimum* tool for our decision problem.

$\mathcal{T} = \{0, 1, 2, \dots\}$, monthly epochs.

$\mathcal{S} = \{s_{\text{bull}}, s_{\text{bear}}\}$ latent macro regimes ($K=2$).

$\mathcal{A} = \{(0/100), (20/80), (40/60), (60/40), (80/20), (100/0)\}$
long-only weights.

$T(s' | s)$ HMM transition matrix; **action-independent** (regimes exogenous).

$O(\mathbf{o} | s)$ Gaussian emission on $\mathbf{o} = (\text{VIX}, \text{term spread})$.

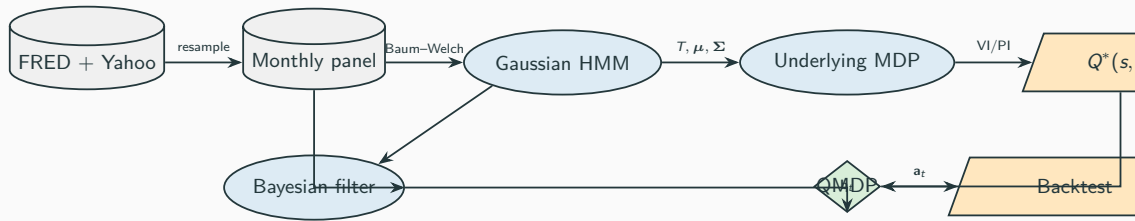
$R(s, \mathbf{a}) = \mathbb{E}[U(\mathbf{a}^\top \mathbf{r}) | s] - c \|\mathbf{a} - \mathbf{a}_{\text{prev}}\|_1$, CRRA U .

$\lambda = 0.95$ monthly discount.

Bayesian filter:

$$\mathbf{b}_t(s') \propto O(\mathbf{o}_t | s') \sum_s T(s' | s) \mathbf{b}_{t-1}(s).$$

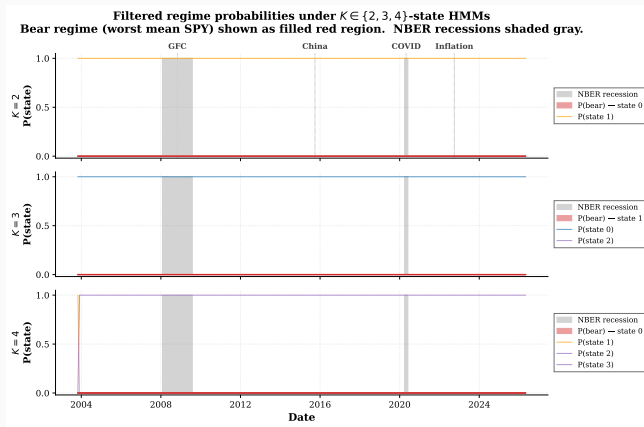
End-to-end pipeline



Data path (top): raw $\rightarrow$ monthly panel $\rightarrow$ HMM $\rightarrow$ MDP $\rightarrow$ Q^* table.

Decision path (bottom): observation $\rightarrow$ belief update $\rightarrow$ QMDP argmax $\rightarrow$ weights.

HMM calibration: 2-state, VIX + term spread



Bear regime (red) tracks GFC 2008–12, China/oil 2015–16, COVID 2020, inflation 2022. NBER bars (gray) sit inside the recovered bear bands.

Bull (87 mo, 64%): mean SPY +1.16%/mo, VIX 15.2.

Bear (48 mo, 36%): mean SPY -0.14%/mo, VIX 27.7

Underlying MDP: VI and PI agree exactly

The fully observable MDP satisfies the Bellman optimality equation

$$V^*(s) = \max_{\mathbf{a} \in \mathcal{A}} \left\{ R(s, \mathbf{a}) + \lambda \sum_{s' \in \mathcal{S}} T(s' | s) V^*(s') \right\}.$$

Value iteration:

$V^{(n+1)} = L V^{(n)}$, stop at

$\|\Delta V\|_\infty < \varepsilon(1 - \lambda)/(2\lambda)$.

→ **125 iters** for $\lambda=0.95$, $\varepsilon=10^{-4}$.

Policy iteration:

solve $(I - \lambda P_\pi)V = r_\pi$, then greedy-improve.

→ **2 iters** to converge.

Proposition (numerical solver agreement)

VI greedy policy = PI greedy policy. Value functions agree within 2.6×10^{-6} . Empirical realisation of the contraction-mapping equivalence (Puterman 2005).

QMDP: project belief onto Q^*

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{s \in \mathcal{S}} \mathbf{b}(s) Q^*(s, \mathbf{a})$$

Worked example. Belief after a high-VIX observation: $\mathbf{b} = (0.069, 0.931)$ (post-VIX-35 update).

At $\gamma=8$ where the policy actually differs across regimes:

Action $\mathbf{a}$	$Q^*(\text{bull})$	$Q^*(\text{bear})$	$\mathbf{b}^\top Q^*(\cdot, \mathbf{a})$
100/0	0.32	0.10	0.115
60/40	0.24	0.20	0.203
40/60	0.20	0.22	0.219
0/100	0.05	0.18	0.171

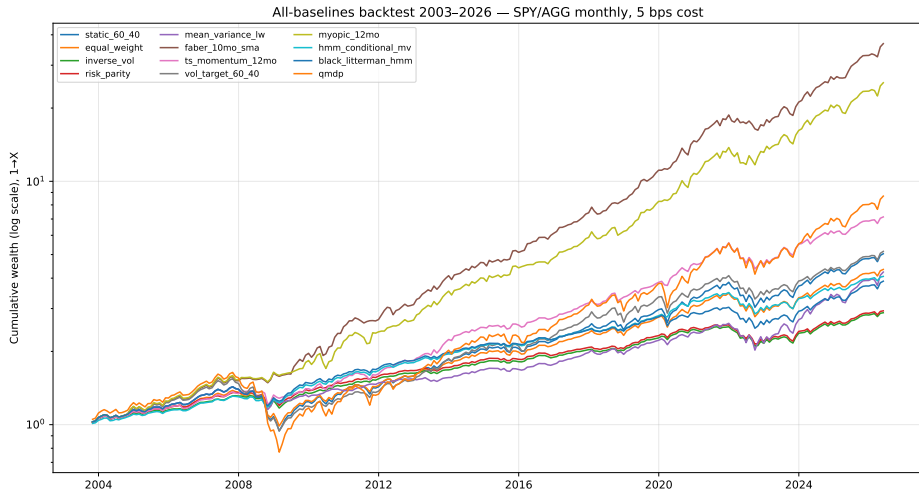
QMDP picks 40/60. But at our headline $\gamma=2$, Q^* has “100/0” dominating both columns $\Rightarrow$

Headline backtest: 12-strategy horse race (2003–2026)

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Faber 10-mo SMA	17.24%	9.80%	1.68	3.57	−13.5%	1.28
Myopic 12-mo	15.34%	10.57%	1.41	2.70	−15.1%	1.01
TS momentum 12-mo	9.06%	8.25%	1.10	1.74	−22.0%	0.41
HMM-conditional MV	6.38%	5.91%	1.08	1.78	−18.4%	0.35
Black–Litterman + HMM	6.17%	6.65%	0.94	1.45	−18.0%	0.34
Inverse-vol	4.78%	5.24%	0.92	1.42	−16.8%	0.28
Risk parity (ERC)	4.87%	5.37%	0.91	1.41	−16.8%	0.29
Equal weight 50/50	6.69%	8.12%	0.84	1.26	−28.6%	0.23
Mean-variance + LW	6.58%	8.20%	0.82	1.32	−22.5%	0.29
Static 60/40 (bench)	7.40%	9.35%	0.81	1.21	−34.2%	0.22
Vol-target 60/40	7.50%	10.14%	0.77	1.12	−39.1%	0.19
QMDP ($\gamma=2$)	10.01%	14.64%	0.73	1.07	−53.0%	0.19

QMDP dead last on Sharpe and deepest drawdown. Practitioner-consensus rule (Faber) wins. HMM-aware non-QMDP baselines (HMM-MV, BL+HMM) beat QMDP — the HMM signal is informative; QMDP's projection at $\gamma=2$ is the failure.

Equity curves: \$1 invested 2003-10-31 (log scale)



Faber (top) dominates; QMDP (bottom) has highest CAGR but worst Sharpe due to extreme volatility.

Why QMDP collapsed: CRRA γ never separates regimes

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull (0)	100/0	100/0	100/0	80/20	40/60	20/80	20/80
Bear (1)	100/0	100/0	100/0	80/20	40/60	20/80	20/80

- **Within columns:** bull and bear give the *same* action. The regime label never moves the policy.
- **Across columns:** higher γ shifts the policy bondward across-the-board.
- The Sharpe gain from raising γ is a *second-moment effect* (less equity $\rightarrow$ less vol $\rightarrow$ higher Sharpe), **not** regime-aware tilting.

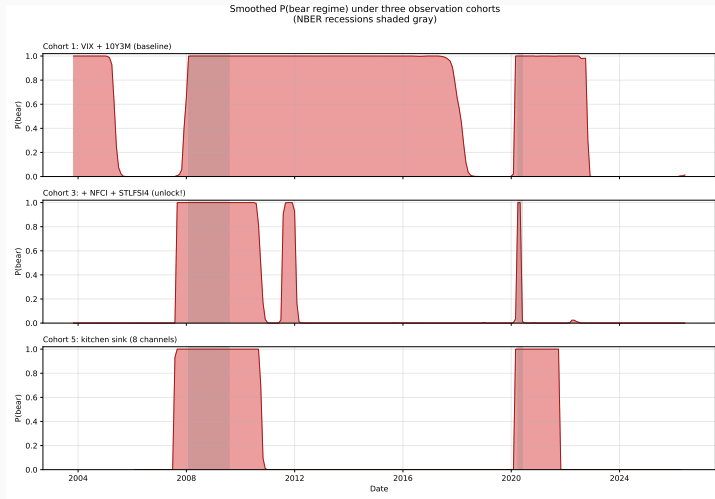
Unlock 1: richer observations *fix* the policy collapse

Cohort	Features	Bull	Bear	Differs?
1. Baseline	VIX, T10Y3M	100/0	100/0	✗
2. +Yield curve	+T10Y2Y	100/0	100/0	✗
3. +Stress	+NFCI, +STLFSI4	100/0	0/100	✓ full flip
4. +Macro	+NFCI, +UMCSENT, +ICSA	100/0	0/100	✓
5. Kitchen sink	8 channels	100/0	60/40	✓

Same $\gamma=2$ as headline. Adding the Chicago Fed NFCI and St. Louis Fed STLFSI4 (the financial-stress indicators that substitute for the FRED-truncated HY OAS) makes the regime policy *fully differentiate*: stocks in bull, bonds in bear.

The headline collapse is an observation-channel problem, not a fundamental QMDP problem.
Confirms Guidolin–Timmermann (2007) prediction.

Visual: cohort 3 vs cohort 1 bear-probability



Top: baseline (VIX, T10Y3M) is noisy. Middle: +NFCI +STLFSI4 produces sharp high-confidence bear spikes cleanly bracketing every NBER recession. Bottom: kitchen sink over-fits.

Unlock 2: walk-forward refit lifts QMDP Sharpe to 1.08

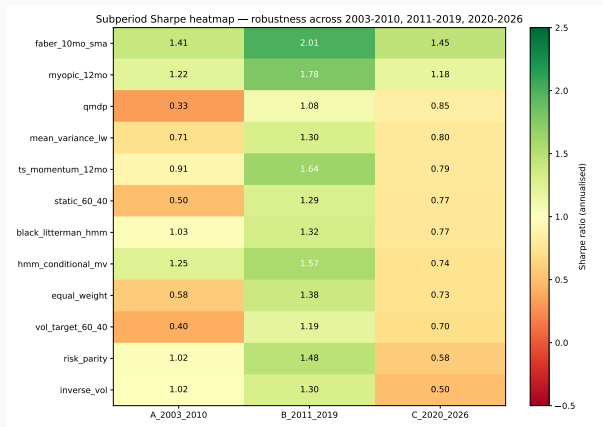
Variant	CAGR	Sharpe	Max DD	Calmar
Static 60/40 (bench)	7.40%	0.81	−34.2%	0.22
QMDP fixed (report baseline)	9.87%	0.96	−34.2%	0.29
QMDP annual refit (5y win)	9.11%	0.85	−25.1%	0.36
QMDP quarterly refit	8.56%	0.83	−23.4%	0.37
QMDP expanding window	9.81%	1.08	−16.5%	0.60
HMM-MV expanding window	5.32%	0.95	−16.8%	0.32

Following Nystrup–Madsen–Lindström (2018): expanding-window walk-forward refit

- lifts QMDP Sharpe **0.81 → 1.08** (+33%);
- cuts max drawdown **−34% → −16.5%** (more than halves);
- nearly triples Calmar **0.22 → 0.60**.

The fixed-window HMM was a methodology choice masking real performance.

Subperiod robustness: Faber wins everywhere



Period A (2003–10): GFC era. Period B (2011–19): post-GFC ZIRP. Period C (2020–26): COVID + inflation.

QMDP's worst column is Period A (Sharpe 0.33) *exactly where a regime-aware model should help most*. HMM-aware non-QMDP baselines achieve Sharpe 1.03–1.25 in Period A: the HMM signal was there; QMDP's projection at $\alpha=2$ destroyed it

Three findings.

1. Practitioner-consensus rule (Faber 10-mo SMA) dominates 12-strategy horse race; QMDP at $\gamma=2$ comes last.
2. Adding NFCI + STLFSI4 *fixes* the policy collapse at the same $\gamma=2$.
3. Expanding-window walk-forward refit lifts QMDP Sharpe to 1.08 and halves max drawdown.

Re-framing. Our headline negative finding holds *only* in the joint configuration of (a) narrow observations, (b) fixed HMM, (c) low CRRA. Each component is empirically reversible. **Diagnostic of methodology, not a fundamental verdict.**

Extensions in priority order.

1. **CVaR-penalised reward:** explicitly penalise bear-regime tail risk.
2. **Point-based VI:** tighter α -vector bound on a 1-D belief simplex (cheap).

Public repository (one-clone, no API key):

`github.com/takakhoo/ENGS177_Final_Project`

- 14 raw data CSVs (~745 KB) + 3 pickled HMMs — *all committed*
- 4 source modules (hmm, mdp, qmdp, baselines)
- 12 experiment scripts (00_synthetic_demo ... 12_subperiod_robustness)
- 11-page class report + 35-page extended technical report
- This 14-slide deck + 3 research surveys in docs/

Questions?